Climate Change Drivers

- Describe environmental and social impacts with IPAT

  Impact = Population · Affluence · Technology

- Carrying Capacity
  - number of people that can be supported indefinitely by the resources of a specific area
- Biocapacity
  - Productivity of Land Areas

- Ecological Footprint
  - measurement of current annual human demand on Earth's ecosystems
- Ecological Deficit
  - difference between population's ecological footprint, and biocapacity of area inhabited by population

- Natural Capital
  - Earth's resources that are used by humans

- Greenhouse Effect Equivalents to CO_2

GHG	GWP
CO_2	1
CH_4	21
N_2O	310
HFCs	124-14800
PFCs	7390-12200
SF_6	22800
NF_3	17200

- Carbon Footprint
  - area of land/air/sea required to absorb equivalent CO_2 [Global Hectares]

- Biodiversity
  - the variety of species and ecosystems on Earth
  1. Ecosystems Diversity
  2. Species Diversity
  3. Genetic Diversity

- Ecosystem Services
  - support from ecosystems on mankind
  1. Air/Water purification
  2. Waste decomposition
  3. Soil/Nutrient cycling
  4. Climate/Radiation regulating
  5. Habitat Preservation
  6. Noise Reduction
  7. Aesthetic/Culture
  8. Raw Materials/Products

- Engineers and Geoscientists of BC Guidelines
  1. Maintain a current Knowledge of Sustainability
  2. Integrate Sustainability into Professional Practice
  3. Collaborate with Peers and Experts on Projects
  4. Develop Clear Justification to Implement Sustainable Solution
  5. Assess and Identify Opportunity for Sustainable Improvement

- Argumentation
  - occurs in social environment
  - communication that aids the consideration of issues and may lead to resolution of issue
  1. Claim
  2. Reason
  3. Evidence
  4. Warrant

Systems Theory
  - a group of interconnected components that work together for common purpose
  System if:
  1. discrete elements/components
  2. interconnected components
  3. different interconnected effect

  - Simple System: whole is equal to sum of components
  - Complex System: whole is greater than sum of components

  - Systems Thinking: being aware of physical structure and scale, context, relationships and processes

- Complexity Science
  Common Properties:
  1. Complex Collective Behaviour: collective actions of vast numbers of components
  2. Signaling and Information Processing: signals and information to/from complex systems
  3. Adaption: complex systems' behaviour changes for improvement

  - Self-Organizing: systems in which organized behaviours form independently
  - Emergent: macroscopic behavior of nearly unpredictable systems

- Resilience
  - protects nested sub-systems from disturbances

  - Adaptive Cycle: describes behaviour of constantly changing complex systems
    1. Rapid Growth: abundant resources facilitate rapid growth
    2. Conservation: scarce resources diminishing growth
    3. Release/Disturbance: disturbance causes quick collapse
    4. Reorganization: quickly reorganize into different structure following collapse

- Concept Map
    - provide context for situation/event

    Characteristics:
    1. Concepts: perceived regularity in events/objects
    2. Linking Words: used to link concepts into a proposition
    3. Concept Hierarchies: general concepts higher up, specific concepts lower down
    4. Cross-Links: links between concepts in different segments or domains of concept map
    5. Examples: specific examples of events/objects to clarify given concept.

    Creating:
    1. Begin with familiar domain of knowledge
    2. Create ranked list of key concepts
    3. Construct a draft concept map
    4. Add cross-links

Sustainable Cities

- cities are complex, thermodynamically open, self-organizing
- cities bring together different socio-economic, natural, professional cultures

- Livability: the quality of interactions between people and their urban environment

    Indicators:
    1. Air Quality
    2. Working Conditions
    3. Sports Activities

- City Resilience Framework
    1. Health and Wellbeing: all citizens have what they need to survive and thrive
    2. Economy and Society: social and financial systems enable urban populations to live peacefully
    3. Leadership and Strategy: promote effective leadership, inclusive decision-making, integrated planning
    4. Infrastructure and Environment: human-made and natural systems that provide critical services

- Ecosystem Approach: protect and manage environment through scientific reasoning

- Integrative Design Process
    - focuses on resource efficiency by employing systems thinking

    Stages:
    1. Design Preparation
    2. Evaluation
    3. Conceptual Design
    4. Schematic Design
    5. Design Development
    6. Construction Documents
    7. Bidding and Construction

    Design Goals:
    1. Energy
    2. Water
    3. Material
    4. Site
    5. Indoor Quality

    - Design Charette: intensive planning session for creating an overarching project vision

- Leadership in Environmental and Energy Designs
    - certification based on sustainable design categories

- Living Building Challenge
    - certification based on sustainable attributes:
        1. Materials
        2. Zero Net Energy
        3. Beauty
        4. Site Integrity
        5. Health
        6. Zero Net Water
        7. Equity

Supply Chains

- economic growth is becoming less dependent on natural resource consumption

- Dematerialization requires
    1. Government Policy
    2. Resource-efficient business models
    3. Better consumer product choice

- Biomimicry: engineering based on natural systems/structures

- 6 Design Principles
    1. Evolve to Survive
    2. Adapt to Changing Conditions
    3. Be Locally Attuned and Responsive
    4. Integrate Development with Growth
    5. Be Resource Efficient
    6. Use Life-Friendly Chemistry

- Industrial Ecology (Symbiosis)
    - a network of industries such that the waste of one is used by another

- Circular Economy
    - minimize input of new materials
    - adopt renewable energy
    - focus on minimizing waste

- Life-Cycle-Assessment Methodology: material flow assessment to determine environmental impact
    - First Phase: Questions
        - Purpose of LCA?
        - Scope of LCA?
        - Functional Units for LCA?

- Second Phase: Inventory Analysis
  - measure throughput of each phase
- Third Phase: Impact Classification
  - each flow identified is categorized into one of several types of impacts
- Fourth Phase: Interpretation
  - organize results to allow for a comprehensive understanding of data

Engineering in Least Developed Countries

- Humanitarian Engineers
  - Altruistic
  - Creative
  - Possess engineering knowledge
- "Wicked" Problems
  - hard to define
  - multiple solutions
  - unique
- Ethical Frameworks for International Engagement
  1. Ethics of Care: non-vulnerable populations should help vulnerable populations
  2. Non-Maleficence & Beneficence: avoid needless harm & promote well-being of others
  3. Ethical Pluralism: theory that there are differing rights & wrongs
  4. Feminism: everyone is equal
- Seven Questions to Sustainability — International Institute for Sustainable Development
  1. Engagement
  2. People
  3. Environment: long-term effectiveness
  4. Economy: economic viability
  5. Traditional & Non-market Activities are maintained
  6. Institutional Arrangements & Governance
  7. Synthesis & Continuous Learning